

1(a)	<i>assume in context of transmission electron micrograph unless otherwise stated</i> <i>any one from:</i> <i>idea that can see internal structures ;</i> <i>cannot see surface contours / AW ;</i> A not 3-D appearance <i>AVP ; e.g. ref. to small(er) depth of field ;</i>	1
1(b)	<i>any two from:</i> <i>(flattened) sacs have layered appearance / no connection between membranes / AW / ora ;</i> <i>not, connected to / contiguous with / continuous with, (outer membrane of) nuclear envelope / ora ;</i> <i>swellings at end of sacs (for vesicle formation) / vesicles at ends of sacs ;</i> <i>no ribosomes / ora ;</i>	2
1(c)	actual diameter = image length / magnification ; <i>at 11 mm for X–Y:</i> 234 nm ; A 213 nm (for 10 mm) A 255 nm (for 12 mm) A 223 nm (for 10.5 mm) A 245 nm (for 11.5 mm)	2

Question	Answer	Marks
1(d)(i)	I glycocalyx I glyco / carbohydrate chains A points from labelled diagram 1 phospholipid <u>bilayer</u> ; <i>plus any two from:</i> 2 fatty acid core / fatty acid tails orientated inwards ; 3 protein ; 4 further detail of proteins ; e.g. integral and peripheral / AW scattered (in phospholipid bilayer) 5 cholesterol ;	3
1(d)(ii)	<i>any one relevant suggestion:</i> reduced gaps between membrane molecules ; higher proportion of phospholipids with saturated fatty acids / ora ; fewer unsaturated fatty acids so, fewer 'kinks' in tails / closer packing ; higher proportion of cholesterol molecules ; fewer, channel / carrier / transport, proteins ; smaller diameter of channels in non-specific channel proteins ; fewer types of (specific), transport / carrier, proteins ; AVP ; e.g. fewer, aquaporins / channels for water	1